THE UNIVERSITY OF BRITISH COLUMBIA
Okanagan Campus

Health Informatics Equity Lab

From AI literacy to AI agents: building preparedness across undergraduate and graduate nursing education

Charlene E. Ronquillo, Ramon Lawrence, Josephine Nanyonjo, et al.
School of Nursing & Department of Computer Science, The University of British Columbia Okanagan, Canada

Background

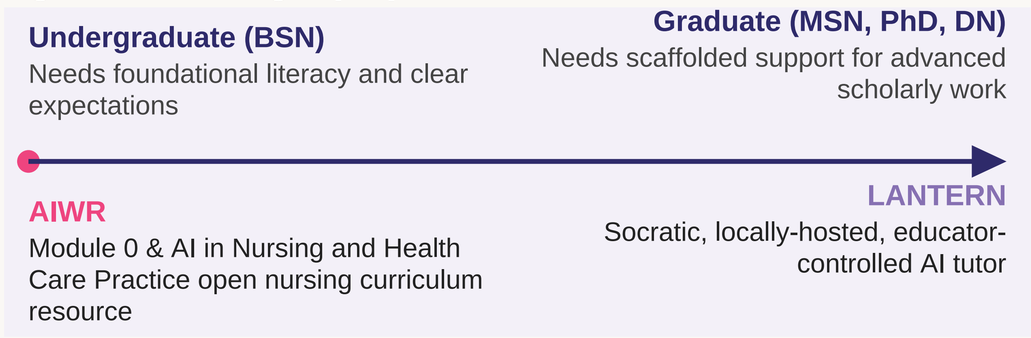
AI is reshaping nursing education, and nursing leaders are calling for responsible, discipline-specific preparation (Ronquillo et al., 2021; Ali & Shaban, 2026).

- Undergraduate BSN students need foundational AI literacy and clear expectations.
- Graduate students are often mid-career nurses who work full time and care for family. Much of their scholarly work happens in the evenings and on weekends, when faculty support is unavailable.
- Coordinated approaches that span the full education continuum remain rare.

A continuum approach

UBC Okanagan School of Nursing runs two complementary initiatives, each matched to its learners. AIWR builds foundational literacy for undergraduate BSN students; LANTERN provides educator-controlled scholarly support for graduate students.

Figure 1. Matching AI preparedness to the learner



UBCO joined the national AI Workforce Readiness (AIWR) consortium, led by the Alberta Machine Intelligence Institute (Amii).

- Faculty piloted Module 0, a 3-hour online foundational AI literacy module in Canvas.
- Faculty and students found it clear, and asked for captions, earlier definitions and an at-a-glance summary. Its upbeat tone may not reach AI-hesitant learners.
- With Bow Valley College, MacEwan University and BCIT, faculty co-developed the AI in Nursing and Healthcare Practices curriculum resource.
- Its four openly licensed sections, each with cases and labs, cover AI foundations in a clinical context, AI-augmented documentation, clinical reasoning with AI, and ethical, legal and professional accountability.

Progress to date

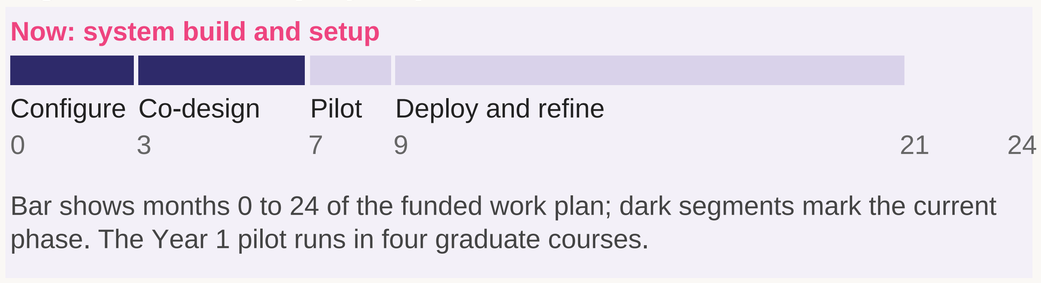
- AIWR: Module 0 now runs nationally as AI for All: Essentials for Students, a free, self-paced course. Locally, pilot feedback prompted faculty talks about threading AI literacy through the BSN.
- Curriculum resource: completed May 2026.
- LANTERN: in system build and setup, co-designed by nursing, computer science and health-system partners. The team is gathering pilot-course materials to build the system and finalizing the ethics submission for its formal evaluation.
- Reach: 40 to 70 students a year across the MSN, PhD and forthcoming Doctor of Nursing programmes.

Graduate: LANTERN

LANTERN (Learning and AI Navigation for Transformative Education and Research in Nursing) is an educator-controlled Socratic AI tutor that asks students how they reached a conclusion and what evidence supports it, instead of giving answers,

- It runs on HelpMe, a UBCO-built platform with a locally hosted language model and no external AI vendor.
- Faculty-verified curriculum materials ground its responses, and faculty review of responses flagged as uncertain by the system.
- LANTERN supports self-regulated learning, which is expected to drive the longer-term outcomes.

Figure 2. LANTERN project phases



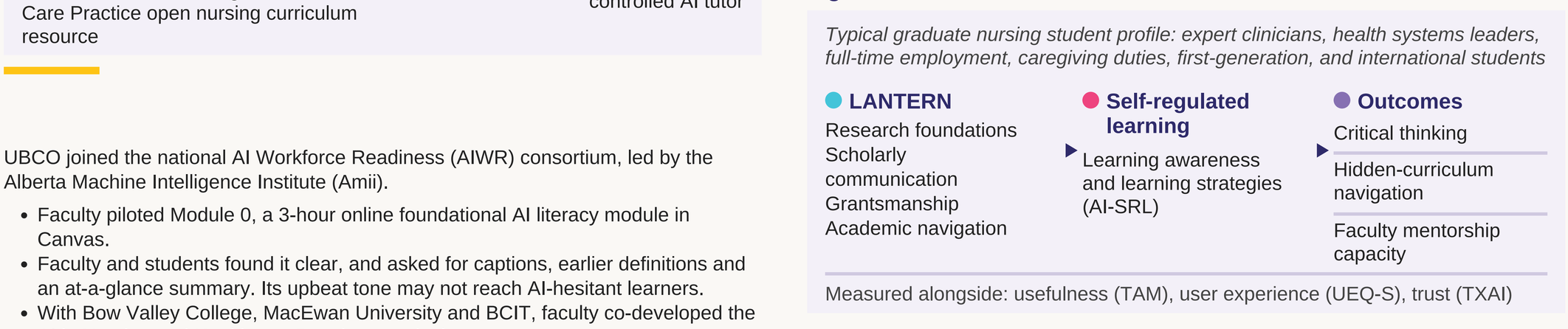
Bar shows months 0 to 24 of the funded work plan; dark segments mark the current phase. The Year 1 pilot runs in four graduate courses.

Evaluating LANTERN

Prospective observational 24-month convergent mixed-methods evaluation with about 30 graduate students and 8 to 10 faculty.

- Students complete surveys at the start, middle and end of term on usefulness, user experience, trust and self-regulated learning, and rate each interaction.
- Platform analytics track use, return visits and escalations to faculty.
- Faculty report on response quality, review workload and mentorship.
- Interviews and focus groups run in two waves, and results are reported by learner group.

Figure 3. How LANTERN works



Lessons and limitations

What we have learned so far

73% of School of Nursing faculty surveyed (April 2026, n = 22) felt only slightly supported or not supported in using AI, citing limited time and training.

- Consult faculty early and often: most surveyed faculty already use AI in teaching, mainly to draft cases and quiz items.
- Collaboration between institutions widens reach and shares the workload.
- Design for equity from the start: students and faculty co-design in rotating cycles, and usage data are reviewed by learner group.

Limitations

- Both initiatives are early, and learning outcomes are not yet evaluated.
- The findings come from a single institution, and small numbers limit subgroup analysis.

Conclusion

Pairing national AI literacy work with an educator-controlled AI agent offers a replicable, model that enables focus on equity factors. It prepares learners at each level of nursing education to use AI responsibly. A human-in-the-loop design responds to documented risks of generative AI in nursing (Topaz et al., 2025).

Authors: Charlene Ronquillo, Ramon Lawrence, Josephine Nanyonjo, Lindsay Burton, Casey Hewes, Jelena Komanchuk, Kathy Rush, Laura Mercer, Charlene Strumpel, Leah Dawe, Lisa Moralejo, Kathryn Douglas-Campbell, Ryan Wilson, Melanie Willson, Dennis Jasper, Sarah Camacho, Clare Harvey.

References

Ali, S. I., & Shaban, M. (2026). AI literacy in nursing education: Building workforce readiness for safe and ethical integration into practice. Nurse Education in Practice, 92, 104771. https://doi.org/10.1016/j.nepr.2026.104771

Ayaz, M. A., Karataş, F., & Yüce, E. (2026). Development and validation of the AI-Enhanced Self-Regulated Learning (AI-SRL) scale. RIED-Revista Iberoamericana de Educación a Distancia, 29(1), 185–216. https://doi.org/10.5944/ried.45452

Ronquillo, C. E., Peltonen, L. M., Pruinelli, L., et al. (2021). Artificial intelligence in nursing: Priorities and opportunities from an international invitational think-tank of the Nursing and Artificial Intelligence Leadership Collaborative. Journal of Advanced Nursing, 77(9), 3707–3717. https://doi.org/10.1111/jan.14855

Topaz, M., Peltonen, L. M., Michalowski, M., et al. (2025). The overlooked dark side of generative AI in nursing: An international think tank's perspective. Journal of Nursing Scholarship, 57(4), 559–562. https://doi.org/10.1111/jnu.70016

Funding: UBC Okanagan ALT2040 Fund; AI Workforce Readiness Program, Alberta Machine Intelligence Institute (Amii). Contact: healthinformatics.equity@ubc.ca